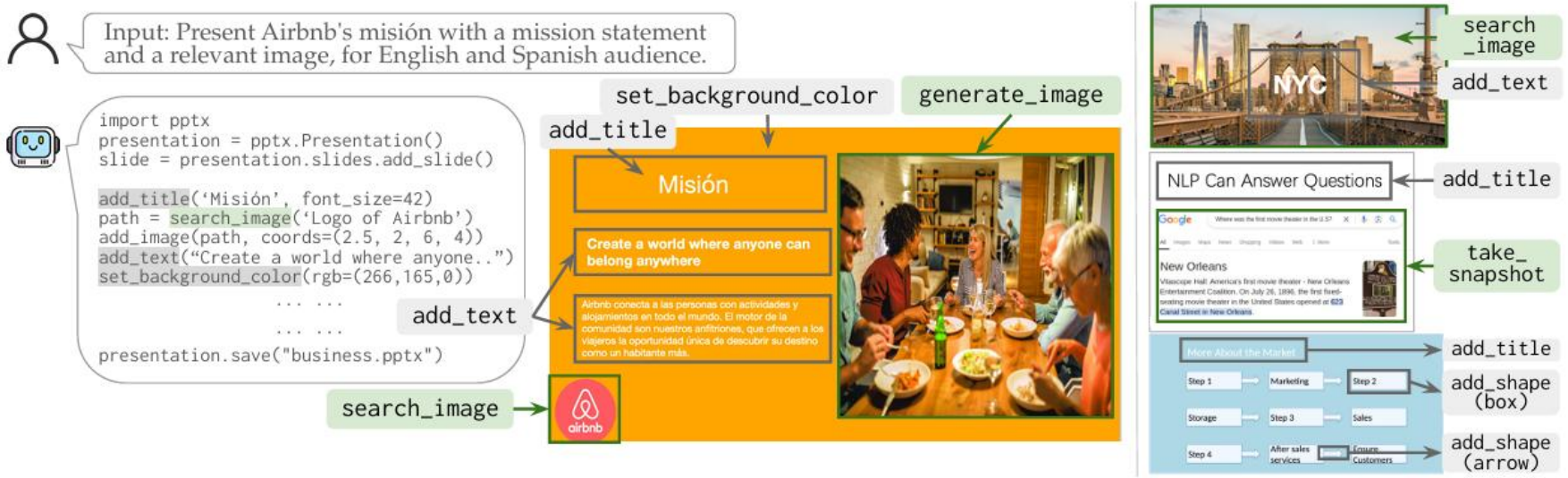
AutoPresent: Designing Structured Visuals from Scratch

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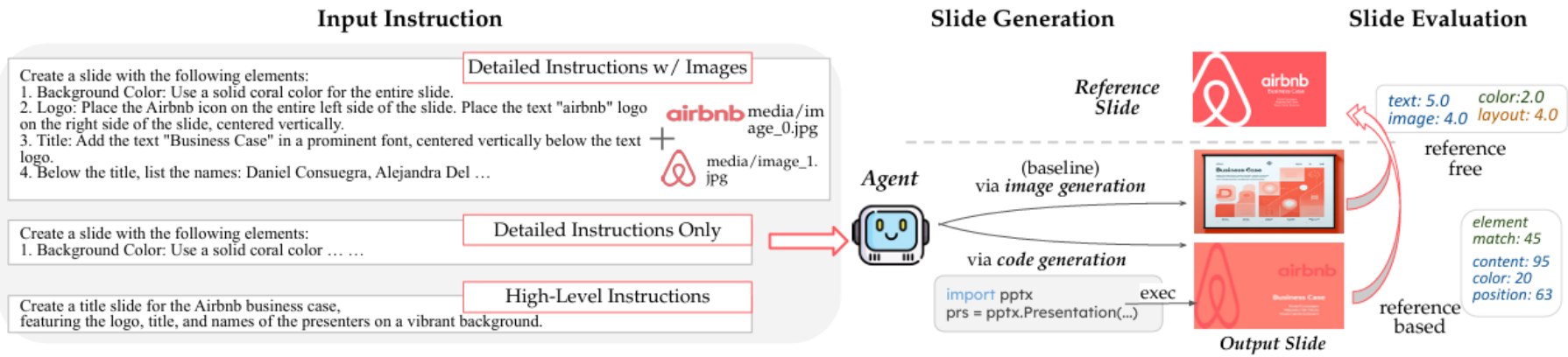
Introduction

- Creating high-quality slides is difficult and time-consuming.
- Existing AI models struggle with structured slide design.
- We define the NL-to-slide generation task and introduce SlidesBench (7k examples).
- We use program generation to create controllable, editable slides, and propose AutoPresent + SlidesLib.



SlidesBench Overview

- Collected 310 high-quality slide decks from 10 domains.
- Built 7k training and 585 test examples (PPTX format).
- Three task setups: detailed+images, detailed-only, high-level.
- Instructions created using human annotation + GPT-4o-mini.



Evaluation Metrics Overview

Three metric types

- Reference-based: element, content, color, position similarity
- Reference-free: text, image, layout, color quality
- Executability: whether generated code runs successfully

Method

- Slides generated via NL-to-code.
- Model outputs Python-PPTX programs from instructions.
- SlidesLib simplifies code: 170 lines -> 13 lines.
- AutoPresent: 8B LLaMA trained on 7k instruction–code pairs.
- Iterative refinement improves slide quality.

Function	Description
`add_title`	Insert a title in the slide.
`add_text`	Insert text at a specific location.
`add_bullet_points`	Insert a textbox with bullet points.
`add_image`	Insert image at a specific location.
`generate_image`	Call an image generator (Dall-E 3) given a query.
`search_image`	Search for an image on a search engine (Bing).
`search_screenshot`	Display a query on a web browser (Google Chrome) and take a snapshot of the search result.

Experiments

- Code Gen Setup: sample 3 programs; first executable is used.
- Baselines: LLAMA, DeepseekCoder, CodeLLaMA, LLAVA, GPT-4o.
- Also compare end-to-end image models: SD2, DALL·E 3.
- Evaluate with reference-based, reference-free, and executability.

Results

Method	Execution%	Reference-Based				Reference-Free				Overall
		element	content	color	position	text	image	layout	color	
Reference	100.0	–				59.7	81.5	73.5	65.7	–
End-to-end Image Generation										
Stable-Diffusion*	**100.0**	74.5	33.4	9.0	75.0	19.6	45.1	36.9	40.5	48.0
DALLE 3*	**100.0**	75.5	39.9	9.2	76.1	32.7	87.3	56.7	53.4	50.2
Code Generation w/o SLIDESLIB										
LLaVA (7B)	11.3	61.9	**97.3**	6.2	70.8	41.6	**100.0**	29.2	25.7	6.1
CodeLLaMA (7B)	5.1	63.6	94.0	11.2	74.0	52.0	43.0	48.0	40.0	3.1
LLaMA (8B)	2.1	74.0	94.6	12.5	**81.2**	50.0	8.3	50.0	50.0	1.3
GPT-4o	89.2	83.3	91.6	10.5	77.0	51.9	72.8	53.7	54.7	55.1
AUTOPRESENT (ours)	79.0	67.7	79.7	10.9	75.9	45.3	62.7	54.2	60.9	45.2
Code Generation w/ SLIDESLIB										
LLaVA (7B)	20.0	80.5	80.5	3.5	64.0	37.5	48.0	29.5	43.5	9.7
CodeLLaMA (7B)	48.7	80.3	89.8	9.4	69.3	45.9	66.8	45.1	49.9	30.3
LLaMA (8B)	54.4	78.3	91.2	7.5	69.5	46.0	68.2	47.6	53.1	33.5
GPT-4o	86.7	**86.2**	92.5	12.7	76.3	**54.6**	83.7	**70.5**	59.4	**58.0**
AUTOPRESENT (ours)	84.1	84.2	92.2	**18.1**	67.2	47.8	73.2	58.6	**64.7**	55.0

Conclusion

- Introduced SlidesBench for evaluating NL-to-slide generation.
- AutoPresent + SlidesLib performs close to GPT-4o.
- Iterative refinement further improves slide quality.

Reference

- J. Ge *et al.*, "AutoPresent: Designing Structured Visuals from Scratch," *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Nashville, TN, USA, 2025, pp. 2902-2911, doi: 10.1109/CVPR52734.2025.00276.